

Rutgers University  
Master's Comprehensive Examination  
Part I      Theory (90 minutes)      December 2, 1989

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

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1) Let  $X$  and  $Y$  be continuous random variables having joint density  $f$  given by  $f(x,y) = \alpha^2 \exp(-\alpha y)$ ,  $0 \leq x \leq y$ , and  $f(x,y) = 0$  elsewhere.

- a) Find the marginal density of  $X$ .
- b) Find the marginal density of  $Y$ .
- d) Find the joint cumulative distribution function of  $X$  and  $Y$ .

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2) Let  $X_1, X_2, \dots, X_n$  be independently and identically distributed random variates with Uniform distribution on  $(0, \theta)$ , where  $\theta > 0$ .

- a) Find the maximum likelihood estimate of  $\theta$ .
- b) Find the exact distribution of the maximum likelihood estimator.
- c) Construct an exact 95% confidence interval for  $\theta$ .

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3) Twenty-five numbers are rounded off to the nearest integer and then added. Assume the individual round-off errors are independent and uniformly distributed over  $(-0.5, 0.5)$ . Find the probability that the given sum will differ from the sum of the original 25 numbers by more than 3.

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4) Let  $X_1, X_2, \dots, X_n$  be a random sample from the density

$$f(x; \theta) = \theta (1+x)^{-(1+\theta)} I_{(0, \infty)}(x), \quad \text{for } \theta > 0.$$

- a) find a complete and sufficient statistic if one exists.
- b) find the Cramer-Rao lower bound for unbiased estimators of  $1/\theta$
- c) Find the UMVUE of  $1/\theta$  if such exists.
- d) Find the UMVUE of  $\theta$  if such exists.

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5) Let  $X$  be a single observation from the density

$$f(x; \theta) = \theta x^{\theta-1} I_{(0,1)}(x), \quad \text{where } \theta > 0.$$

- a) In testing the Null Hypothesis:  $H_0: \theta \leq 1$ , versus the Alternative Hypothesis  $H_1: \theta > 1$ , find the power function and size of the test given by the following; Reject  $H_0$  if and only if  $X \geq 0.5$ .
  - b) Find a most powerful size alpha test of  $H_0: \theta = 2$ , versus  $H_1: \theta = 1$ .
  - c) Is there a uniformly most powerful size alpha test of  $H_0: \theta \geq 2$  versus  $H_1: \theta < 2$ ? If so what is it?
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PART II

Rutgers University  
Master's Comprehensive Examination  
Applied (90 minutes)

December

INstructions: Attempt any four out of the following five questions. Mention which four you would like to have graded. Open book and open notes.

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1) In a large legislative body, 50% of the members are age 35 to 55, 25% are age 56 to 64, and 25% are age 65 or older. In a sample of 20 people chosen to form a key committee, 5 people are age 35-55, 8 people are age 56-64, and 7 people are age 65 or older. It is claimed that the 20 people were chosen completely at random from the legislative body. Carry out a test of this claim.

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2) Let  $P$  denote the probability that a spun penny will come up tails. We seek to test the Null Hypothesis:  $P = 0.5$ , versus the Alternative Hypothesis  $P > 0.5$ . We are going to spin the penny 100 times and observe  $X$ , the total number of tails in the 100 spins. The test will reject the Null Hypothesis if  $X$  exceeds some critical number  $C$ .

a) Find  $C$  such that the test will have a 5% level of significance.

b) For the value of  $C$  in part (a), find the Power of the test against the alternative  $P = 0.60$ .

c) If  $X = 60$ , what would be the observed level of significance ( $p$ -value)?

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3) A researcher is investigating the relation between the amount of money spent on probation services (per offender placed on probation) by a county, and the total number of crimes committed in those counties. For a sample of 20 counties, the mean amount spent on probation services is \$1,000, with a standard deviation of \$500. For these same counties, the mean number of crimes committed is 100 (per community), with a standard deviation of 30 crimes. The correlation between number of crimes committed and the amount spent on probation services is  $-0.60$ . (a negative correlation).

a) The least-squares estimate of the average number of crimes committed in counties that spend \$1800 on probation is

b) The estimate of the standard deviation of the number of crimes committed in communities that spend \$1800 on probation per offender is

c) The 95% confidence interval for the average number of crimes committed in communities that spend \$1800 on probation is

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4) Suppose that we have a chemical process in which a feed stock is pretreated in one of several ways to form an intermediate, and then the intermediate is processed in a second stage at one of two specific temperatures. Suppose there are 4 specific methods of pretreatment and 5 batches are prepared by each method. From each of these 20 batches we take 12 samples and treat 6 at one temperature and the other 6 at the other temperature. The goal of the study is to determine how long the feed stock will remain good under the different pretreatment-temperature processes. It is realized that each batch cannot be pretreated under ideal controlled conditions.

- (a) Identify the response variable and the different factors.
- (b) For each factor, state whether it is a treatment or a blocking factor, and whether it is fixed or random. Briefly justify your answer.
- (c) For each pair of factors, state whether they are crossed or nested. Draw a data display.
- (d) Write the full model for this example which accounts for all the possible effects. Clearly identify the indices.

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 5) A researcher seeks to test a new reducing diet on 10 women selected at random with the results given in the table below.

| Woman | Weight before | Weight after |
|-------|---------------|--------------|
| 1     | 140           | 132          |
| 2     | 145           | 140          |
| 3     | 156           | 155          |
| 4     | 158           | 155          |
| 5     | 162           | 158          |
| 6     | 189           | 192          |
| 7     | 195           | 185          |
| 8     | 210           | 186          |
| 9     | 264           | 238          |
| 10    | 322           | 160          |

The researcher carries out a t-test on this data.

- a) Explain why the researcher's choice of test is inappropriate here.
- b) Carry out an appropriate test to determine whether the diet helps to reduce weight. Can you conclude (at the 5% level of significance) that the diet is effective?
- c) Give a one-sided 80% confidence interval for the average weight loss of the form: the average weight loss is at least \_\_\_\_\_ pounds. State which average you are using.

Do Not Give Out

Rutgers University  
Department of Statistics  
Master's Comprehensive Re-Exam

Part I

Theory (90 minutes)

March 25, 1989

Instructions: Open book and open notes.

1. Let  $X$  and  $Y$  be two independent exponential random variables with common mean 2. Find  $E(\frac{1}{X+Y})$ .

2. Let  $X$  and  $Y$  be two independent normal random variables such that

$$EX = \mu, \quad EY = \mu, \quad \text{Var}(X) = 3, \quad \text{Var}(Y) = 2.$$

Find the maximum likelihood estimation (MLE) of  $\mu$  based on  $X$  and  $Y$ . Show that the MLE is also unbiased.

3. Let  $X$  be a normal random variable with  $EX = \mu$  and  $\text{Var}(X) = 1$ . Let

$$Y = e^{tX^2}$$

for some  $t < 1/4$ . Find the mean and variance of  $Y$ .

4. A coin is tossed 8 times. Assume that

$$P\{\text{Head}\} = 1/3 = 1 - P\{\text{Tail}\}.$$

Find the probability of having at least four consecutive tails.

Rutgers University  
Department of Statistics  
Master's Comprehensive Re-Exam

Part II

Applied (90 minutes)

March 25, 1989

Instructions: Open book and open notes.

1. A college professor wishes to know how well a final exam mark can be predicted from a midterm exam mark. The following table gives marks for five students.

|                  |    |    |    |    |    |
|------------------|----|----|----|----|----|
| Midterm mark (X) | 81 | 73 | 77 | 66 | 55 |
| Midterm mark (Y) | 88 | 80 | 84 | 87 | 69 |

Fit a straight line for Y in terms of X and test the hypothesis that the slope of the line equals 1.

2. In a study of smoking habits, 1000 persons were asked whether they were nonsmokers, moderate smokers or heavy smokers. The results were broken down by sex.

|       | Nonsmokers | Moderate smokers | Heavy smokers | Total |
|-------|------------|------------------|---------------|-------|
| Men   | 96         | 81               | 103           | 280   |
| Women | 335        | 210              | 175           | 720   |
| Total | 431        | 291              | 278           | 1000  |

- (a) Is there any evidence that smoking habits differ between men and women?
- (b) Estimate the proportion of nonsmoking females in the population and give the mean and variance of your estimate.
3. A new reducing diet was tested on nine women selected at random, with the results in the table below.
- (a) Give an 89% confidence interval for the mean weight difference in the population represented by the nine women.
- (b) At a 1% significance level, would you conclude that the diet is effective?

| <u>Woman</u> | <u>Weight before</u> | <u>Weight after</u> |
|--------------|----------------------|---------------------|
| 1            | 133                  | 132                 |
| 2            | 146                  | 143                 |
| 3            | 135                  | 137                 |
| 4            | 168                  | 160                 |
| 5            | 148                  | 151                 |
| 6            | 152                  | 148                 |
| 7            | 180                  | 171                 |
| 8            | 126                  | 120                 |
| 9            | 122                  | 124                 |

## Master's Comprehensive Re-Exam/Part II/Applied/3-25-89.

4. In order to study the efficiency of its staff, a local bank manager decided to observe the traffic in the bank's lobby for an hour. During this period ten customers arrived, of which only seven left before the hour was over. Their individual service times were recorded as 10, 13, 12, 8, 30, 6 and 5 minutes. For the three customers who did not complete their activities at the end of the observation period, the manager recorded only that their individual service times EXCEEDED 3, 7 and 4 minutes.

Under the assumption that the service times are independent random variables with a common exponential distribution ( $F(x) = 1 - e^{-\theta x}$ ,  $x > 0$ ,  $\theta > 0$ ), what should be the manager's estimate of the mean and standard deviation of the service time, if he maximizes the likelihood of the data.